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Modulation of Tropical Cyclogenesis by ITCZ characteristics in the Southern Indian Ocean

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Background and Aim

- **Tropical Cyclones** (TC) tend to form close to the Intertropical Convergence Zone (ITCZ) due to the presence of abundant moisture, instability and low-level vorticity.
- Idealized simulations have shown a linear relationship between the ITCZ location, TCs and their precursors ("seeds")[1]. In the observations, the ITCZ location modulates TCs in the Northern Hemisphere, but not in the Southern Indian Ocean (SIO)[2], where predictability is more limited compared to other regions.
- Moreover, on interannual time scales, a relevant factor for tropical cyclogenesis (TCG) is the meridional gradient of zonal wind[3]. Periods of a strong ITCZ are hypothesized to be associated with a more cyclonic gradient and frequent wave breaking.

Does the ITCZ location and strength modulate TCG on interannual scales in the SIO?

Data and Methods

- **Seed dataset[4]**: Tracking on ERA5 using geopotential height at 850 hPa if the first point occurs between 25N and 25S and where relative humidity > 75%. Seeds are defined as developers if the track matches the reference IBTrACS TC track.
- **Survival rate** is defined as $SR = {}^{TC}/_{SD} = {}^{TC}/_{TC+ND}$, where SD is the number of Seeds, while ND the number of Non-Developers.
- **Period**: 1980/81 – 2020/21 (41 seasons). Months: November to April (SH TC Season).
- **ITCZ Metrics**: Computed using 6-hourly ERA5 in the box (0-20S, 30-135E) following the precipitation centroid method.

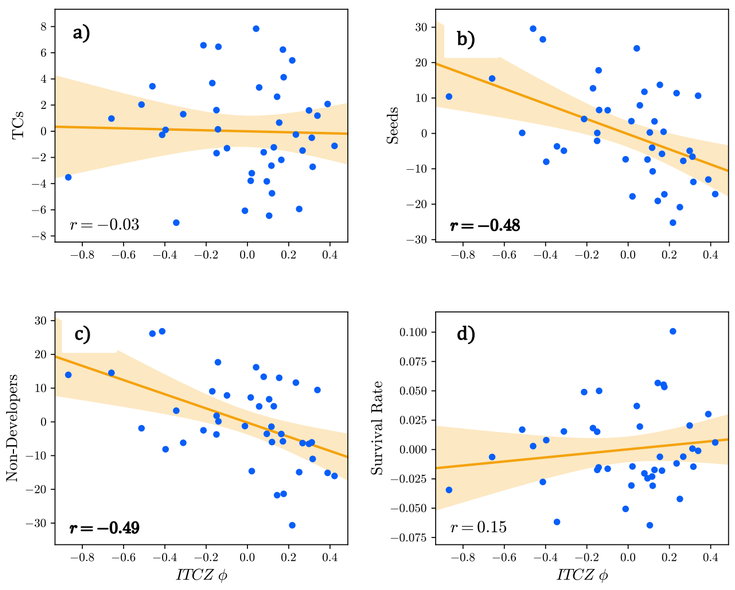
Location

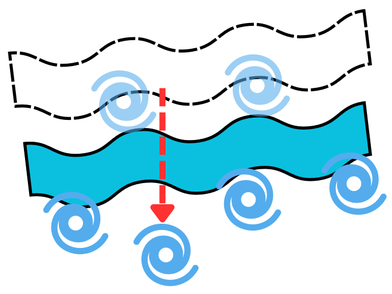
$$\phi_{ITCZ} = \frac{1}{N_\theta} \sum_{\theta=30}^{135} \frac{\int_{-20}^{0} p(\phi,\theta,t)\ \phi\ \cos\phi\ d\phi}{\int_{-20}^{0} p(\phi,\theta,t)\ \cos\phi\ d\phi}$$

Strength

$$p_{ITCZ} = \frac{1}{\theta_2 - \theta_1} \int_{\theta_1}^{\theta_2} p(\theta', \phi_{ITCZ})\ d\theta'$$

TCG and ITCZ Location





More Non-Developers as the ITCZ shifts poleward

Fig.1 – Scatterplots of (a) TCs, (b) Seeds, (c) Non-Developers, (d) Survival Rate against ITCZ location ϕ. Lines show the linear fit with [95%] confidence interval. Bottom-left text indicates the Pearson correlation coefficient, with bold values denoting significance ($p < 0.05$).

TC interannual variability

$$N'_{TC} = \overline{N}_{Seed} \times SR' + \overline{SR} \times N'_{Seed} + \epsilon$$

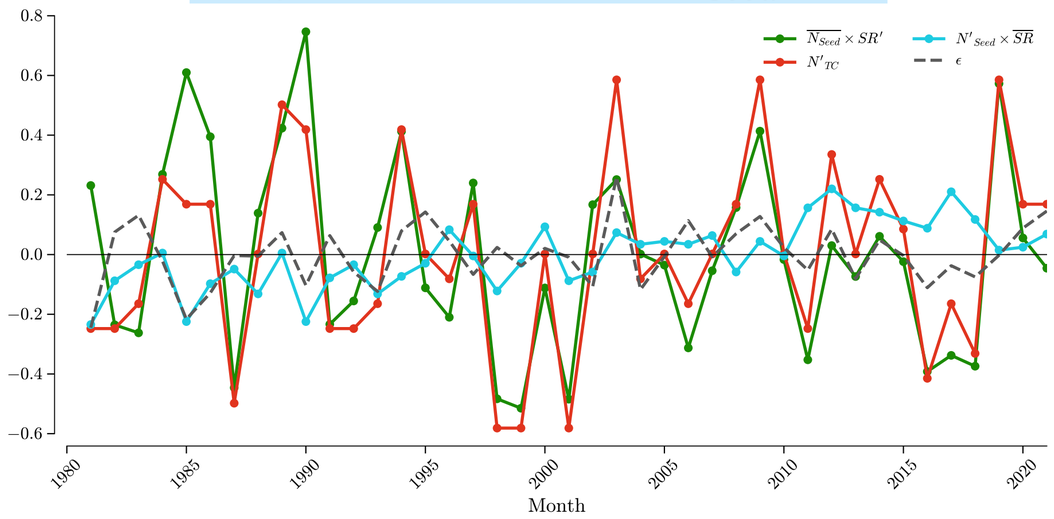
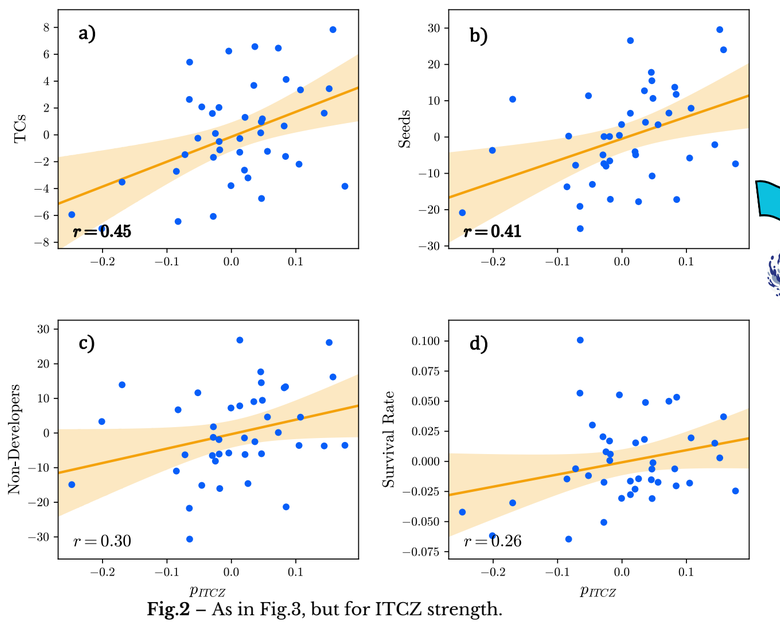


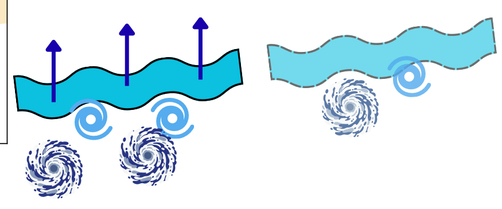
Fig.2 – Decomposition of the interannual variability of TCs (red) in the contribution of survival rate (green), seeds (blue) and residual (black). ' indicates an anomaly relative to the (1980-2021) climatology.

On interannual time scales, TC variability is mainly driven by changes in Survival Rate

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TCG and ITCZ Strength





More seeds and TCs as the ITCZ strengthens

Fig.2 – As in Fig.3, but for ITCZ strength.

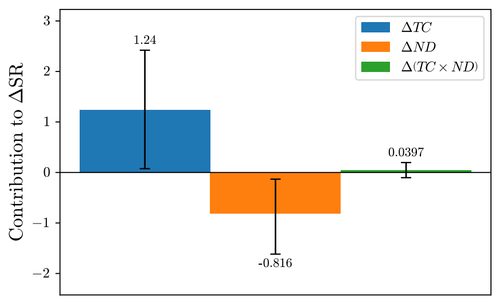
If TC variability is mainly determined by changes in the SR, why do only TCs and seeds respond to variation in ITCZ strength?

Changes due to Non-Developers Variability

$$\Delta SR = \Delta TC\ \overline{ND} + \Delta ND\ \overline{TC} + \Delta(\mathrm{TC} \times ND)$$

Changes due to TC variability | **Covariance term**

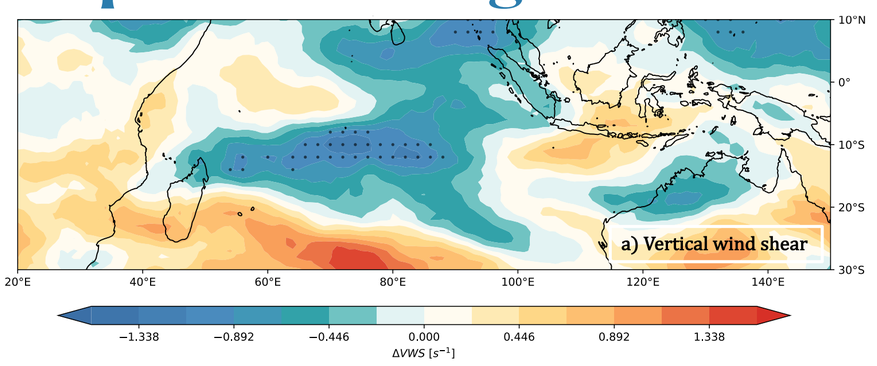
$$\Delta SR = SR\big|_{strong\ ITCZ} - SR\big|_{weak\ ITCZ}$$

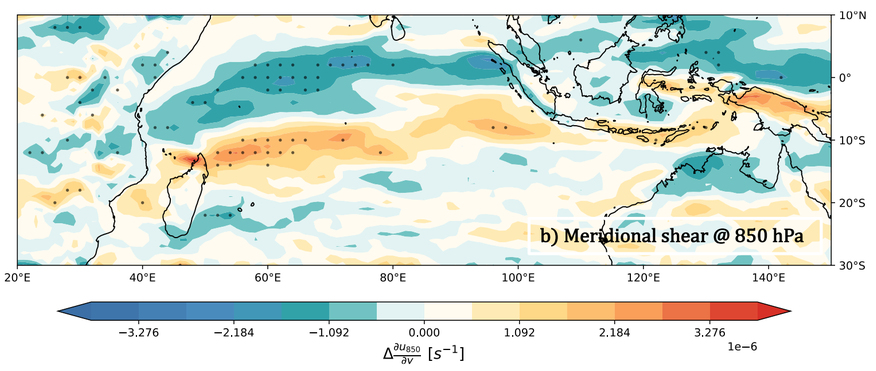


Competing effect of TCs and Non-Developers on Survival Rate variability

Fig.4 – Bar plot of the contributions to the seasonal mean difference in survival rate (SR) during seasons of different ITCZ strength. Error bars represent the 2.5- 97.5 percentile range derived from a 1,000-bootstrap of annual data. Values are computed over the Southern Indian Ocean basin (0-30S, 30-135).

Composites: Strong – Weak ITCZ





More favourable environment for seed initiation and intensification

Fig.5 – Differences between the top 12 years of strong ITCZ and the 12 weak ITCZ in the anomalies of vertical wind shear between 200 hPa and 850 hPa and (b) meridional shear of the zonal wind at 850 hPa. Stippling indicates statistical significance at the 95% level.

Summary & Next Steps

- In the SIO, the ITCZ location modulates seed frequency only, while its strength affects both TC and seed interannual variability. Representation of the ITCZ may benefit TC S2S forecasts in this basin.
- The survival rate should be interpreted with care: it does not capture large-scale conditions becoming more favourable for genesis, due to the competing effect between TCs and Non-Developers.
- How does the interaction between the ITCZ and TCs change on shorter time scales? Do the upper ocean conditions influence the survival rate of seeds? A km-scale regional coupled model is in development to address these questions.

References

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3. Wang, Bin, and Hiroyuki Murakami. "Dynamic genesis potential index for diagnosing present-day and future global tropical cyclone genesis." *Environmental Research Letters* 15.11 (2020): 114008.
4. Moon, Jihong, et al. "Tropical cyclone seed disturbances in ERA5." *Journal of Climate* 38.18 (2025): 4625-4639.